

Standard Operating Procedure: Maintenance & Inspection Protocol for M102

1. Inspection Schedules & Frequencies

Daily Inspection: Check coolant fluid levels via sight gauge, monitor chip conveyor cleanliness, and verify pneumatic pressure (minimum 6.0 bar).

Weekly Inspection: Inspect way-lube oil reservoir, clean telescopic way-cover seals, and test emergency stop circuits.

The recommended cooling-system inspection interval is every 200 operational hours (or monthly, whichever occurs first). This inspection must verify radiator airflow, clean heat exchange coils, check pump impellers, and sample coolant pH levels (optimal 8.0 - 9.5).

2. Corrective Action for Cooling Failure & Overheating

Step 1: Isolate power and tag-out Machine M102 per Lock-Out Tag-Out (LOTO) protocols.

Step 2: Inspect radiator intake filter mesh for metal chip shavings and dust buildup. Vacuum or clean with solvent.

Step 3: Check coolant pump flow rate. If flow rate is below 15 L/min at 3.0 bar, replace coolant filter element (Part No. FLT-M102-25).

Step 4: Execute a thermal calibration test run for 30 minutes at 6,000 RPM before resuming full production load.

3. Maintenance Request Protocol

Whenever thermal shutdown occurs or cooling performance degrades, technicians must submit a High-Priority Maintenance Request immediately so that plant engineering can dispatch a specialized technician prior to resuming batch operations.